MNIST

Pattern Recognition - Baseline Experiment

This experiment demonstrates supervised learning using topological associative memory. Raw images have been preprocessed with Gaussian blur.
Image data is encoded as sparse distributed representations (SDRs), using a random sparse projection followed by k-winners-take-all. We encode labels as non-overlapping random sparse holographic representations (SHRs).

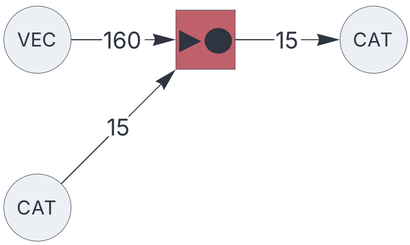
Requirements

```
<< "Circuits.m"
```

Circuit

```
{
  "$schema": "https://creatingintelligence.org/schemas/circuit-v1.json",

  "hyperparameters": {"12": [16000, 160], "default": [150, 15]},

  "dataflow": [
    {"component": "input", "plugin": "category", "send": [11], "categories": 10},

    {"component": "input", "plugin": "vectorencoder", "send": [12]},

    {"component": "associator", "send": [101], "receive": [{"slot": 11, "tags": ["show_population"]},
{"slot": 12, "tags": ["show_population"]}]},

    {"component": "output", "plugin": "category", "receive": [{"slot": 101, "tags": ["show_population"]}],
"categories": 10}
  ]
}
```

```
circuit = Circuit[json];
c = circuit["function"];
circuit["schematics"][ImageSize→ 220]
```



Dataset

```
labeleddata = Import[NotebookDirectory[] <> "/data/MNIST-blurred.wxf"];
Length /@ %
```

```
{70 000, 70 000}
```

Train

```
{data, categories} = Take[#, 60000] & /@ labeleddata;
```

```
c @@@ Transpose[ {categories, data}]; (* Train *)
```

```
results = c[Null, #]& /@ data; (* Test with training data *)
```

```
100. Mean@Boole@MapThread[SameQ, {results, categories}] (* Score *)
```

```
99.9983
```

Test

```
{data, categories} = Take[#, -10000] & /@ labeleddata;
```

```
results = c[Null, #]& /@ data; (* Test with test data *)
```

```
100. Mean@Boole@MapThread[SameQ, {results, categories}] (* Score *)
```

```
96.04
```